PRODUCT DESIGN SPECIFICATION

Ask Me Loader Experience

Final guidelines for progress communication, phase behavior, typography, color, motion, errors, interactions, responsive layouts, and accessibility.

Version
1.0

Platform
Desktop and mobile web

Primary typeface
Inter

Technical typeface
JetBrains Mono

1. Experience objective

The loader should make long-running AI and data operations feel transparent, predictable, responsive, trustworthy, and easy to inspect without overwhelming the user.

> **The loader must answer:** What is happening? What has completed? What is active now? What happens next? Does the user need to take action?

Avoid generic messages such as “Loading…” when meaningful progress information is available.

2. Loader anatomy

Collapsed state

- Status or animated loader icon
- Current phase title
- Elapsed time
- Expand/collapse chevron
- Stop-generation control while active

Example: Charting the distribution… 0:09

Expanded state

- Completed and active phases
- Phase narration
- Agent or task rows
- Input/output cards
- SQL or JSON cards
- Retry and error details

The loader is collapsed by default. Detailed technical content remains secondary to the primary progress message.

3. Standard phase model

Phase	Active label	Completed label	Duration
1	Understanding your request	Understood your request	3.4s
2	Running data queries	Data queries completed	6.0s
3	Charting the distribution	Distribution charted	5.4s
4	Generating insights	Insights generated	8.8s

Total reference duration: approximately 23.6 seconds.

- Only one phase is active at a time.
- Completed phases remain available in the expanded view.
- Active labels use present-progressive language; completed labels use past tense.

- Do not show exact percentages unless progress is measurable.
- Do not delay a ready response to finish an animation.

4. Status states

Pending

Muted icon and text. No animation and no success or error color.

Active

Animated glyph, heading shimmer, present-progressive copy, blue accent, and visible elapsed time.

Complete

Stable checkmark, no animation, and concise past-tense copy. Text can remain neutral.

Retrying

State that recovery is happening. Preserve the original failure in expanded details.

Failed

Terracotta-red icon and heading, clear explanation, and actionable recovery guidance.

Stopped

Stop timers and transitions immediately. Show “Generation stopped” with restart or retry.

5. Error flow

Event	Time
Understanding begins	0s
Understanding completes	3.4s
First warehouse error	6.0s
Automatic retry	8.8s
Hard failure	11.8s
Error experience settles	12.6s

> **Recommended final error:**
> We couldn’t complete the data query after retrying. Check the data source connection or try again.

- Explain what failed, whether the system retried, and what the user can do next.
- Keep error codes and raw provider responses inside expanded details.
- Stop all progress animations after a hard failure.
- Never hide the final failure inside the collapsed state.

6. Typography

Inter, with system-ui and sans-serif fallbacks, is used for headings, phases, narration, controls, and error messages. **JetBrains Mono**, with monospace fallback, is used for timers, duration pills, input/output values, SQL, JSON, and technical metadata.

Element	Size	Weight	Line height	Color
Current loader heading	14px	600	Normal	Slate 800
Phase title	14px	600	Normal	Slate 800
Agent/task heading	14px	500	1.4	Slate 800
Phase narration	14px	400	1.7	Slate 600
Agent supporting detail	12px	400	1.5	Slate 500
Elapsed time / duration	12px	400 mono	Normal	Slate 600
Input/output value	12px	400 mono	1.55	Slate 600
Input/output label	10px	500 mono	1.5	Slate 400
SQL/JSON label	12px	500	Normal	Slate 600
SQL/JSON code	12px	400 mono	1.85	Slate 800
Error heading	12px	600	Normal	Error red
Error metadata label/value	10px / 12px	400 mono	Normal	Slate 400 / 600
Copy confirmation	10px	500	1.4	White

- Never use less than 12px for readable explanations.
- Use 10px only for short metadata or temporary tooltips.
- Keep important phase titles at 14px on desktop and mobile.

7. Color system

Token	Hex	Usage
Slate 800	#1E293B	Primary headings, phase titles, names, code
Slate 600	#475569	Narration, timers, technical values
Slate 500	#64748B	Supporting details and controls
Slate 400	#94A3B8	Metadata and pending indicators
Slate 200	#E2E8F0	Borders and dividers
Slate 100	#F1F5F9	Pills and hover surfaces
Slate 50	#F8FAFC	Code-card background
Blue 600	#2563EB	Active loader glyph
Error red	#B4472B	Failed state text and icons
Success green	#3F7B58	Copy confirmation and success
Terracotta	#C96442	SQL keywords
Chart blue	#3B5E92	SQL identifiers

Use text and icons together. Never rely on color alone to communicate status.

8. Active heading shimmer

```
linear-gradient(90deg, #94a3b8, #1e293b 45%, #94a3b8 90%)
```

- 14px, weight 600
- 2.6-second horizontal movement
- Apply only to the active loader heading
- Do not apply to completed or failed phases

9. Icons

Icon type	Size
Loader status icon	16px
Phase icon	16px
Primary loader glyph	18px
Agent indicator	11px
Composer action icon	20px
Chevron	10–16px

Pair status icons with text. Use blue for active processing, muted slate for pending states, and terracotta red for failures. Completed icons remain stable and non-animated.

10. Motion

Interaction	Duration
Expand/collapse	160–320ms
Content fade-up	200–240ms
Active status pulse	1.6s
Loader bars	1.8s
Loader dots	2.2s
Loader lines	2.4s
Heading shimmer	2.6s
Copy feedback	180ms

- Animate only the active phase.
- Avoid flashing, rapid movement, and competing animations.
- Stop processing animation after completion, failure, or cancellation.
- Respect `prefers-reduced-motion`.

11. Spacing and layout

Area	Specification
Loader header	10px horizontal gap; 8px top and 6px bottom padding
Phase row	9px horizontal gap; 6px vertical padding
Phase body	1px Slate 200 left border; 18px indentation; 12px vertical gap
Agent rows	9px horizontal gap; 9–12px vertical gap; 17px nested indentation
Technical cards	560px max width; 10px radius; 10–14px padding

12. Input and output cards

Labels: JetBrains Mono, 10px, weight 500, Slate 400, letter-spacing 0.06em.

Values: JetBrains Mono, 12px, line-height 1.55, Slate 600. Long values wrap safely. Use Error red only for the failed value or message.

13. SQL and JSON cards

- Slate 50 background and Slate 200 border
- 10px border radius and 12px × 14px padding
- 560px maximum width
- JetBrains Mono at 12px with 1.85 line height
- Default code: Slate 800; keywords: Terracotta; identifiers: Chart blue; strings: Success green

14. Interaction guidelines

Expand and collapse

Make the entire header clickable. Use a stateful chevron, `aria-expanded`, and a connected content region. Preserve expanded state during phase updates.

Stop generation

Show only while active. Use a clear stop icon with the label “Stop generation.” Stop timers, transitions, and pending operations immediately.

Copy

Copy only code content. Show “Copied” for approximately two seconds and provide visual and accessible confirmation.

Replay

Reset elapsed time, active phase, errors, copy feedback, and expanded state before starting the full sequence again.

15. Content guidelines

Active labels

- Understanding your request
- Running data queries
- Charting the distribution
- Generating insights

Completed labels

- Understood your request
- Data queries completed
- Distribution charted
- Insights generated

Use a direct, calm, specific, and technically informative tone. Keep phase labels to 2–5 words and narration to one or two short sentences.

16. Responsive guidelines

Desktop

- Align with the conversation column
- Keep technical cards below 560px
- Place timer and chevron on the right
- Avoid covering the composer

Mobile

- Keep the same typography scale
- Use 44×44px touch targets
- Stack input/output vertically
- Allow code overflow scrolling
- Never depend on hover

17. Accessibility

- Use semantic buttons and accessible labels for icon-only controls.
- Announce meaningful phase changes with `aria-live="polite"`.
- Do not announce timer updates every second.
- Maintain at least 4.5:1 contrast for normal text.
- Ensure keyboard access to expand, stop, retry, replay, and copy.
- Keep focus indicators visible and respect reduced-motion preferences.
- Use empty alt text for decorative images.

> **Recommended announcement:** “Running data queries.” Avoid announcing every timer update or decorative animation.

18. Design tokens

```
:root {
  --loader-text-primary: #1e293b;
  --loader-text-secondary: #475569;
  --loader-text-tertiary: #64748b;
  --loader-text-muted: #94a3b8;
  --loader-border: #e2e8f0;
  --loader-surface-muted: #f1f5f9;
  --loader-surface-code: #f8fafc;
  --loader-active: #2563eb;
  --loader-error: #b4472b;
  --loader-error-border: #e8c4b8;
  --loader-error-background: #fdf7f5;
  --loader-success: #3f7b58;
  --loader-title-size: 14px;
  --loader-body-size: 14px;
  --loader-secondary-size: 12px;
  --loader-metadata-size: 10px;
  --loader-title-weight: 600;
  --loader-agent-weight: 500;
  --loader-body-weight: 400;
  --loader-card-radius: 10px;
  --loader-control-radius: 6px;
  --loader-motion-ease: cubic-bezier(0.4, 0, 0.2, 1);
}
```

19. Do and don’t

Do

- Explain real system activity
- Show meaningful phase transitions
- Keep the default state compact
- Allow technical inspection
- Distinguish retry from failure
- Make all controls functional

Don’t

- Use fake percentages
- Animate every phase
- Hide important failures
- Show raw errors as primary copy
- Use red for normal progress
- Leave animation running after failure

20. Final quality checklist

- [ ] Loader starts only when needed
- [ ] Current phase is understandable
- [ ] Only one phase is active
- [ ] Completed copy uses past tense
- [ ] Elapsed time is accurate
- [ ] Expand works by keyboard
- [ ] Stop affects underlying behavior

- [ ] Replay resets all state
- [ ] Copy provides confirmation
- [ ] SQL and JSON are readable
- [ ] Retry is clearly communicated
- [ ] Failures stop animation
- [ ] Mobile touch targets are sufficient
- [ ] Text contrast is compliant
- [ ] Reduced motion is supported
- [ ] Deployed icons and assets load